
Research article

Safe Active Discrimination of Fractional, Delayed, and Finite Latent Memory in a Strong-Allee Predator–Prey Model

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Abstract: We study how controlled ecological experiments can distinguish Caputo memory from integer-order, delayed, and finite latent alternatives in a strong-Allee predator–prey system. The ecological backbone is intentionally inherited from our previously submitted companion paper: the same strong-Allee model, locked parameters, coexistence equilibrium, and Jacobian are reused as the experimental test bed and are explicitly not claimed as new contributions here. The main text is organized around four analytical results and an expanded numerical study. Structurally, the collocated Caputo transfer has a noninteger high-frequency signature that cannot be exactly reproduced by the declared finite latent or retarded-delay classes. Practically, finite exponential mixtures approximate fractional memory over finite horizons, producing a complexity-dependent testing obstruction. Directly on the ecological prey channel, interval arithmetic certifies finite-state approximations whose error decreases from 0.5335 at $m = 4$ to 0.0070 at $m = 32$ for $\alpha = 0.85$, $A = 0.25$, and $T = 12$; under the declared pulse budget, the associated Gaussian testing lower bound approaches chance. We then introduce a combined fractional-delay model that reduces to the pure Caputo case at $\tau = 0$ and to the DDE case at $\alpha = 1$. New simulations show that delay systematically shifts the predator peak, erodes the Allee safety margin, and creates intermediate responses that are not well represented by either pure limit. Across six perturbation families and a large four-class benchmark, broadband inputs improve discrimination but cross the Allee threshold more frequently, whereas transient inputs are safer numerically but less informative. The results expose a fundamental safety–informativeness trade-off in experimental identification of ecological memory.

Keywords: fractional-order dynamics; time-delay systems; model discrimination; experimental design; Allee effect; predator–prey dynamics

Mathematics Subject Classification: 34A08; 34K20; 62K05; 92D25

1. Introduction

Fractional derivatives are widely used to represent hereditary and nonlocal effects in dynamical systems [4, 6, 11, 16], including predator–prey models [1, 10, 17, 18]. The central empirical difficulty is that a fitted fractional order does not by itself identify a fractional mechanism: over finite horizons, delayed feedback and finite-dimensional latent states can reproduce similar trajectories. Experiment design for fractional-order identification already exists [8, 9, 13, 20], while safety-constrained information acquisition and safe active model discrimination are established general problems [15, 19]. We therefore focus on a narrower question:

Which perturbations and measurements can distinguish fractional memory from delayed or latent alternatives in a strong-Allee ecological system, and when does ecological safety make that discrimination fundamentally difficult?

The ecological backbone is deliberately inherited from our previous validated study [3, 5, 12, 21], so that the present work can isolate a different scientific question: *experimental discrimination of memory mechanisms*. The previous paper established the equilibrium/stability structure, the Matignon boundary, extinction and boundedness certificates, and the failure of a naive Euler surrogate. None of those results is claimed as a contribution here. In the present paper they serve only to define the operating regime and the ecological safety geometry. The new objects are the competing ODE/Caputo/delay/latent response laws, controlled perturbations and observation channels, finite-horizon discrimination bounds, the combined fractional-delay stress test, and the numerical experiment-design benchmark. The ODE and Caputo mechanisms share the inherited ecological equilibrium and Jacobian; retarded-delay and finite-state latent models are calibrated rival response laws compared through matched inputs, observations, and physical units.

Explicit relationship to the previously submitted paper. Because the two manuscripts belong to the same research sequence, the reuse boundary is stated explicitly rather than left implicit. The present manuscript uses the *same ecological backbone*: the strong-Allee predator–prey equations, the locked ecological parameters, the coexistence equilibrium, and the corresponding Jacobian are inherited from the previously submitted companion paper [12]. Those ingredients are not reintroduced as new results and are not part of the novelty claim. They provide a controlled test bed on which a different problem is posed: how to design perturbations and observations that discriminate between alternative memory laws. The distinction is summarized in Table 1.

Table 1. Relationship between the previously submitted companion paper and the present manuscript. The ecological backbone is intentionally shared; the research questions, analytical targets, simulations, and headline results are different.

Aspect	Previously submitted paper [12]	Present paper
Scientific question	Certify equilibria, stability, the Matignon boundary, extinction/boundedness, and discretization-induced stability inversion.	Design and evaluate experiments that distinguish Caputo, delayed, integer-order, and finite-latent memory mechanisms.
Shared material	Strong-Allee ecological equations, locked parameters, coexistence equilibrium, and Jacobian.	The same ecological backbone is reused only as the operating and safety test bed.
Primary mathematics	Validated numerics, Krawczyk/interval certificates, Matignon classification, global extinction/boundedness statements.	Structural transfer separation, finite-horizon latent approximation, testing obstruction, and finite-state prey-response approximation.
Primary computation	Certified parameter grid and certificate inventory.	Frequency-response diagnostics, input-design experiments, four-class model-discrimination benchmark, and fractional-delay simulations.
Headline output	Certified dynamical regimes of the Caputo ecological model.	Safety–informativeness limits for active discrimination of competing memory mechanisms.

This separation is also important methodologically. A stability certificate answers whether a specified Caputo model is locally or globally consistent with a dynamical regime; it does not answer whether the Caputo mechanism can be distinguished from an observationally similar delay or latent-state mechanism. Conversely, the present discrimination results do not recertify the ecological backbone. They condition on the previously established operating geometry and ask what additional perturbation/measurement information is required to tell the candidate memory laws apart.

Main contributions. The paper is organized around four contributions, rather than around a long sequence of standalone theorems.

- 1. Structural and finite-horizon separation.** We show that the collocated Caputo transfer has a noninteger high-frequency signature that cannot be exactly reproduced by finite rational latent models or by the declared retarded-delay class. At the same time, finite exponential mixtures approximate the fractional kernel on a finite horizon, creating a complexity-dependent discrimination barrier.
- 2. Certified prey-response testing obstruction.** Directly on the linearized ecological prey channel, a pole/branch decomposition yields interval-enclosed finite-state approximations. Under a declared pulse budget and Gaussian observations, the corresponding minimax testing lower bound approaches chance as latent complexity grows.
- 3. Numerical validation and a new fractional-delay stress test.** In addition to the calibrated ODE, Caputo, DDE, and latent benchmark, we introduce a combined fractional-delay model. It continuously connects the pure Caputo limit ($\tau = 0$) to the DDE limit ($\alpha = 1$), allowing direct simulation of how fractional order and delay jointly alter phase, transient amplitude, safety margin, and experimental discriminability.

4. **Safety–informativeness trade-off.** Across six input families, broadband excitation improves memory discrimination but increases the rate of crossing the strong-Allee threshold. Transient inputs are safer numerically but less informative, making the constrained experiment-design problem visible rather than merely theoretical.

The detailed proofs, secondary analytical results, Bayesian sequential extension, and solver derivations are moved to the Supplementary Material. The main paper retains only the analytical statements required to interpret the numerical results.

Paper architecture: compact theory, expanded simulations, and a dedicated fractional-delay study

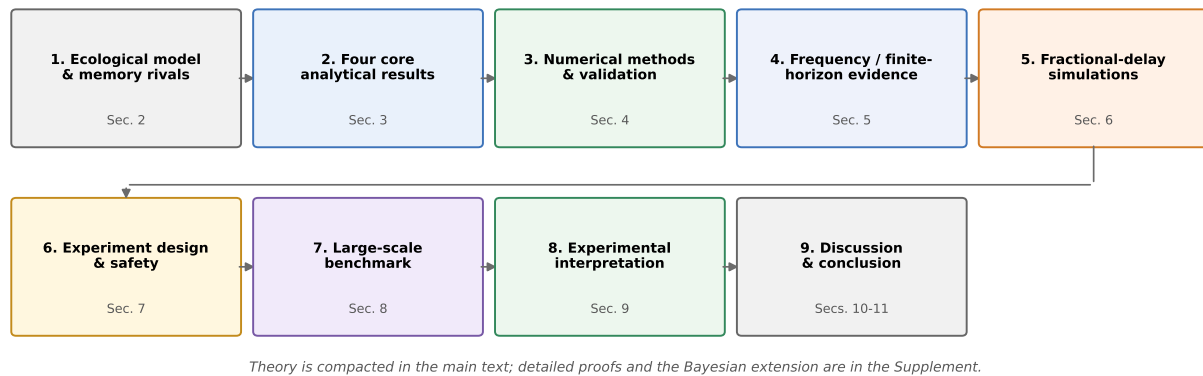


Figure 1. Paper structure after the Q1-focused restructuring: certified ecological model and rival memory laws, four essential analytical results, numerical validation, dedicated fractional-delay simulations, experiment design and safety analysis, and the large-scale discrimination benchmark. Detailed proofs and secondary theory are placed in the Supplementary Material.

2. Ecological Model and Competing Memory Mechanisms

2.1. Strong-Allee predator–prey backbone

This subsection records the ecological backbone inherited from the previously submitted validated study [12]. The model, locked parameterization, coexistence equilibrium, Jacobian, and Matignon stability boundary below are reused inputs rather than new results of the present paper; their derivation and certification belong to Ref. [12]. We restate only the formulas needed to define the controlled discrimination experiments.

The nonlinear ecological model is

$$f_1(x, y) = rx \left(1 - \frac{x}{K}\right) \left(\frac{x}{A} - 1\right) - \frac{axy}{1 + hx}, \quad (2.1)$$

$$f_2(x, y) = e \frac{axy}{1 + hx} - my, \quad (2.2)$$

with locked parameters

$$r = \frac{3}{2}, \quad K = 1, \quad a = 1, \quad h = \frac{1}{2}, \quad e = \frac{4}{5}, \quad m = \frac{2}{5}.$$

The coexistence equilibrium and Jacobian are

$$x^* = \frac{2}{3}, \quad y^*(A) = \frac{2(2 - 3A)}{9A},$$

$$J(A) = \begin{pmatrix} \frac{7A - 2}{8A} & -\frac{1}{2} \\ \frac{2 - 3A}{10A} & 0 \end{pmatrix}.$$

For a commensurate Caputo model, local stability follows the Matignon sector condition [14]. In the complex-pair regime $2/7 < A < 10/19$, the critical order is

$$\alpha^*(A) = \frac{2}{\pi} \arctan \left(\frac{\sqrt{4D(A) - T(A)^2}}{T(A)} \right),$$

with $T(A) = \text{tr } J$ and $D(A) = \det J$. At $A = 0.30$, $\alpha^* \approx 0.969012276$, so the fractional system with $\alpha = 0.9$ is stable although the integer-order limit is unstable [12]. For most discrimination experiments we use $A \leq 2/7$, where the common operating point remains in the stable regime.

Inherited backbone versus new experimental layer. The notation above is intentionally identical to the companion paper because changing the ecological equations would confound the memory-discrimination question with a change of biological mechanism. What changes in the present work is the *response law around the same operating ecology*: the fractional derivative, explicit delay, or finite latent state determines how perturbations propagate through time. This lets all candidates be compared under the same perturbation amplitude, observation horizon, and Allee threshold. Table 2 records the common quantities that are inherited and held fixed unless a numerical sweep explicitly varies A or α .

Table 2. Locked ecological backbone used by both manuscripts. These values are inherited inputs, not new contributions of the present paper.

Quantity	Value	Role in present experiments
r	1.5	intrinsic prey growth
K	1	prey carrying capacity
a	1	attack-rate scale
h	0.5	Holling-II handling parameter
e	0.8	conversion efficiency
m	0.4	predator mortality
x^*	$2/3$	common coexistence prey level
$y^*(A)$	$2(2 - 3A)/(9A)$	coexistence predator level
$J(A)$	Eq. above	ODE/Caputo local ecological generator

The strong-Allee threshold plays two roles in the current manuscript. First, it defines a physically meaningful failure event: if an informative perturbation drives prey below A , the experiment may leave the local coexistence regime and trigger extinction dynamics. Second, the distance $x^* - A$ acts as a natural safety margin. Consequently, an input can be statistically attractive while being ecologically unacceptable, which is the central tension explored numerically in Sections 6, 7, and 8.

2.2. Four rival mechanisms

Dimensional consistency is enforced with the reference time τ_0 [6, 11, 16]. Around the operating point, the four candidate mechanisms are

$$\begin{aligned}
 \text{ODE: } \dot{\xi} &= J\xi + Bu, \\
 \text{Caputo: } \tau_0^{\alpha-1} D_t^\alpha \xi &= J\xi + Bu, \\
 \text{DDE: } \dot{\xi} &= A_0 \xi(t) + A_1 \xi(t - \tau) + Bu, \\
 \text{latent: } \dot{q} &= \bar{A}q + \bar{B}u, \quad y = \bar{C}q.
 \end{aligned}$$

Only the Caputo mechanism carries a fractional order. The finite latent class is restricted by an explicit state-dimension budget m .

Combined fractional-delay stress-test model. To address the interaction of memory and explicit delay directly, the numerical study also uses

$$\tau_0^{\alpha-1} D_t^\alpha x(t) = P(x(t); A) - \frac{ax(t)y(t)}{1 + hx(t)} + u(t), \quad (2.3)$$

$$\tau_0^{\alpha-1} D_t^\alpha y(t) = e \frac{ax(t - \tau)y(t)}{1 + hx(t - \tau)} - my(t). \quad (2.4)$$

A constant equilibrium history is used on $[-\tau, 0]$. Equation (2.4) reduces to the pure Caputo model at $\tau = 0$ and to the retarded DDE benchmark at $\alpha = 1$. It is used as a numerical stress test, not as an additional headline theorem.

System, observation channels, and rival memory laws

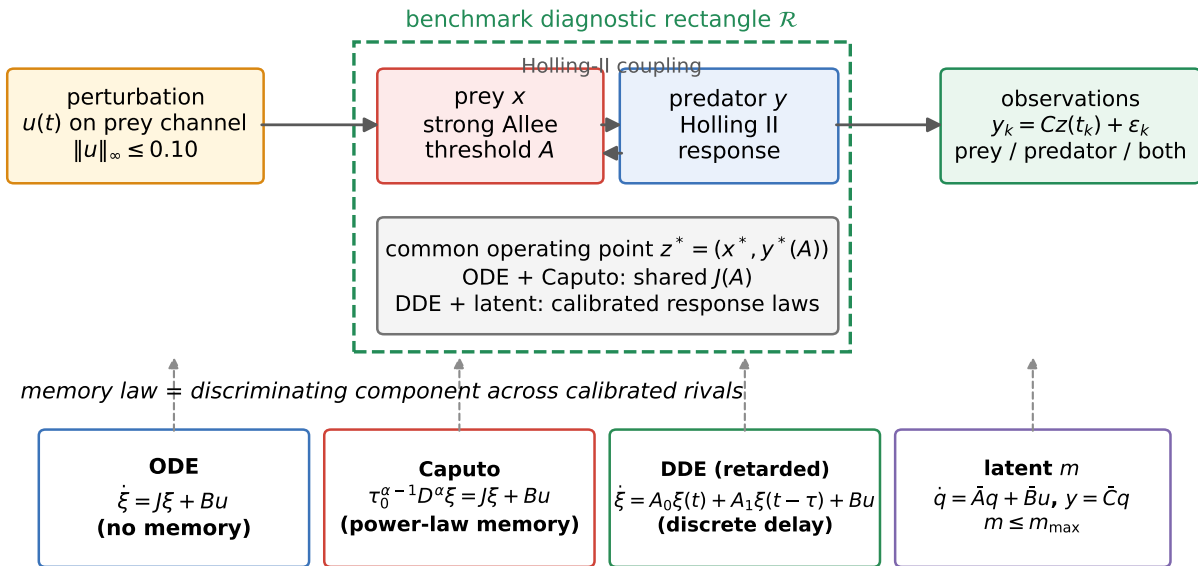


Figure 2. Ecological and experimental setting. The strong-Allee backbone and its parameterization are inherited from the previous validated study [12]; the intervention/observation architecture and the competing memory laws are specific to the present work. The system is perturbed on the prey channel and observed through prey, predator, or both populations. ODE and Caputo models share the exact ecological Jacobian; DDE and latent models are calibrated rival response laws. The dashed rectangle is a numerical benchmark diagnostic region, not an invariant-set certificate.

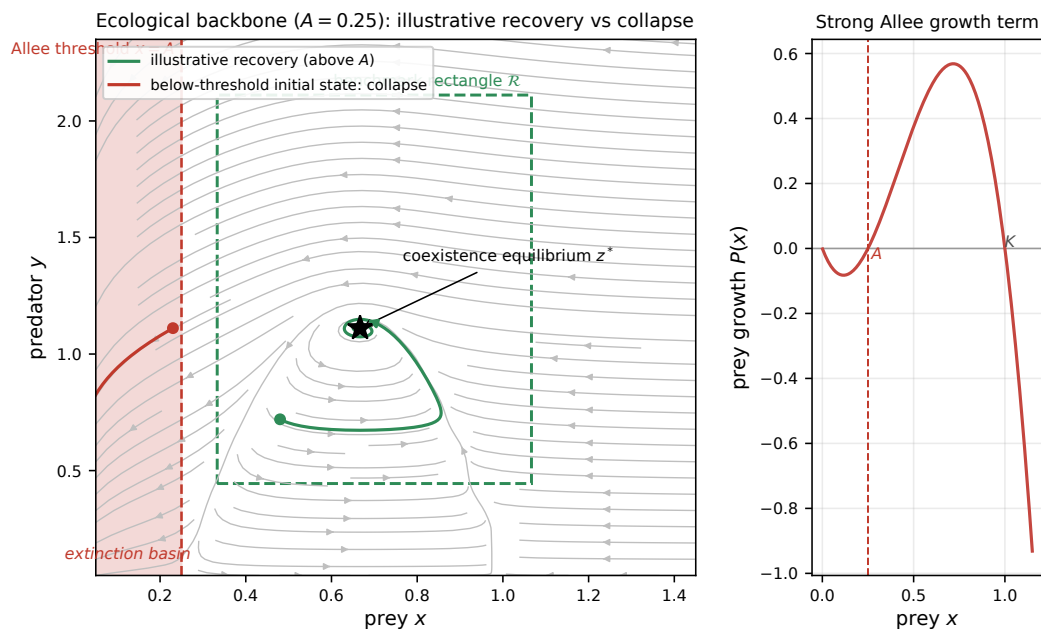


Figure 3. Inherited strong-Allee backbone [12] at $A = 0.25$, shown here only to define the safety geometry for the new experiment-design problem. Left: representative recovery above the threshold and collapse from below A . Right: prey growth term $P(x)$ showing the sign change at the Allee threshold.

3. Analytical Backbone: Four Results Used by the Experiments

The complete derivations and secondary results are given in the Supplementary Material. Here we retain only the four statements that directly support the numerical study.

3.1. Result I: structural separation

For the collocated channel $CB \neq 0$, the Caputo transfer

$$G_\alpha(s) = C(q_\alpha(s)I - J)^{-1}B, \quad q_\alpha(s) = \tau_0^{-1}(s\tau_0)^\alpha,$$

has the high-frequency expansion

$$G_\alpha(s) = (CB)\tau_0^{1-\alpha}s^{-\alpha} + O(|s|^{-2\alpha}).$$

Theorem 3.1 (Structural separation). *For $0 < \alpha < 1$ and $CB \neq 0$, the Caputo transfer cannot equal any finite-dimensional rational latent transfer on a nonempty open subset of a common analytic domain. It also cannot equal a strictly proper finite-state retarded model with finitely many discrete delays, because the latter has $O(\omega^{-1})$ high-frequency decay whereas the Caputo channel has $O(\omega^{-\alpha})$ decay.*

The theorem explains why exact discrimination is possible in principle. It does not imply that finite noisy experiments are easy.

3.2. Result II: finite-horizon latent approximation

The Caputo memory kernel admits a diffusive representation and can be approximated on $[0, T]$ by a positive exponential sum. For a constructive m -mode approximation k_m ,

$$\|k_\alpha - k_m\|_{L^1(0, T)} \leq \widehat{E}_m, \quad \widehat{E}_m \downarrow 0$$

along an expanding nested rate hierarchy.

Theorem 3.2 (Finite-horizon approximation and complexity barrier). *For every $T < \infty$ and $\varepsilon > 0$, there exists a finite positive exponential mixture whose $L^1(0, T)$ distance from the Caputo kernel is below ε . Consequently, without a finite complexity restriction on the latent rival, no fixed finite-horizon experiment can guarantee a uniform positive separation from the entire latent hierarchy.*

This is the analytical reason for treating latent dimension as an experimental-model complexity budget rather than an unconstrained nuisance parameter.

3.3. Result III: testing obstruction under a physical input budget

Let S be the observation operator and R the Gaussian covariance. Define

$$C_{\text{obs}} = \frac{1}{2}\|R^{-1/2}S\|^2.$$

Young's inequality converts the approximation bound into an observation-space bound.

Theorem 3.3 (Complexity-dependent testing obstruction). *If the admissible experiment class satisfies $\|u\|_2 \leq B$, then the composite minimax error for testing the fractional response against a latent hierarchy containing an explicit competitor with error $\widehat{E}_m$ obeys*

$$P_e^*(m) \geq \Psi\left(C_{\text{obs}}\widehat{E}_m^2B^2\right),$$

where Ψ is any valid equal-covariance Gaussian two-point lower bound (Pinsker is used in the numerical plots). Thus $P_e^*(m) \rightarrow 1/2$ whenever $\widehat{E}_m \rightarrow 0$ and B remains bounded.

3.4. Result IV: ecological prey-response approximation

The ecological result works with the exact prey-to-prey fractional transfer, avoiding the invalid factorization $(s^\alpha I - J)^{-1} = (sI - J)^{-1}s^{-\alpha}$. A pole/branch decomposition realizes the principal-sheet pole pair exactly with a stable two-state integer-order block and approximates only the branch remainder.

Theorem 3.4 (Certified finite-state approximation of the prey response). *For the displayed Matignon-stable cells and $m \in \{4, 8, 16, 32\}$, there exists an explicit stable finite-state prey-response surrogate g_m such that*

$$\|g_\alpha - g_m\|_{L^1(0,T)} \leq \widehat{E}_m^{\text{state}},$$

where the bound is obtained by outward-rounded interval subdivision and a rigorous tail estimate. At $\alpha = 0.85$, $T = 12$, the certified enclosure at $A = 0.25$ decreases from 0.5335 ($m = 4$) to 0.0070 ($m = 32$).

For the declared pulse, $\|u\|_\infty \leq 0.10$ implies $\|u\|_2 \leq 0.120$. At $\sigma = 0.10$, the resulting Pinsker testing lower bounds at $A = 0.25$ are approximately 0.340, 0.462, 0.492, and 0.498 for $m = 4, 8, 16$, and 32, respectively. Hence the finite-state rival becomes statistically indistinguishable from the fractional prey response as its allowed complexity grows.

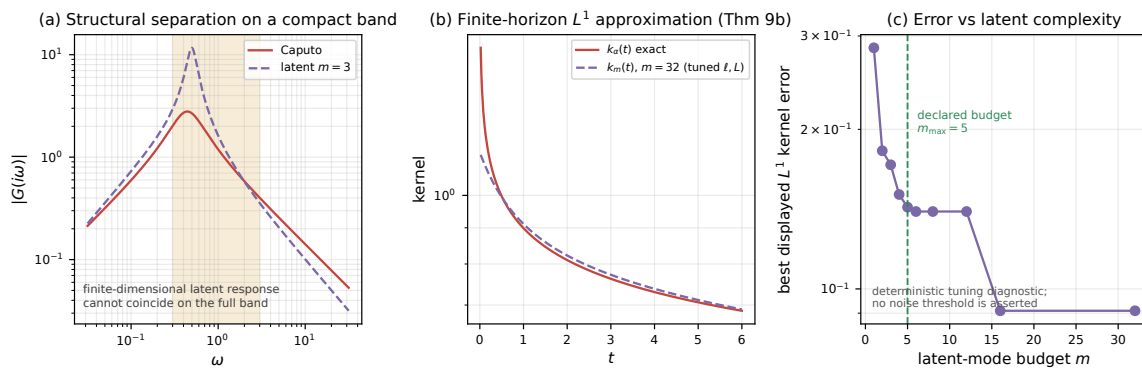


Figure 4. Exact structural separation and finite-horizon approximation are complementary, not contradictory. The Caputo transfer has a distinct noninteger signature, yet increasingly rich finite exponential mixtures track the fractional kernel over a finite horizon.

Certified prey-response finite-state approximation and testing bound ($\alpha = 0.85$, $T = 12$)

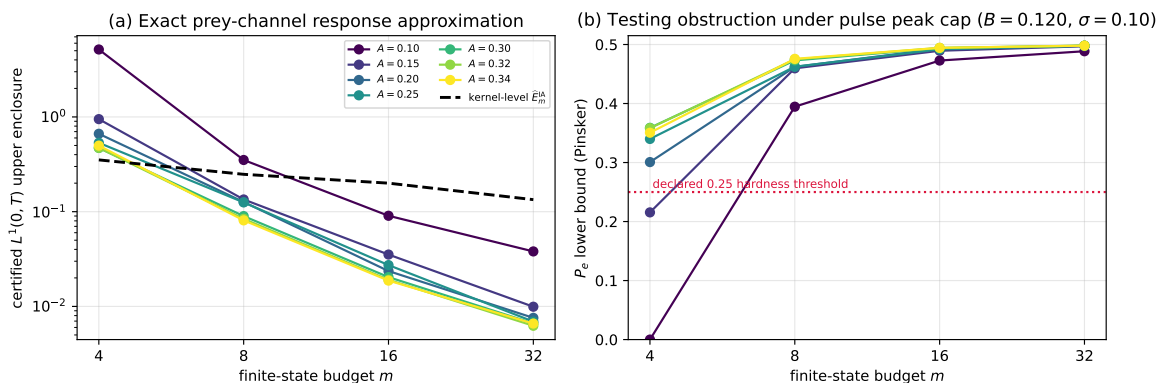


Figure 5. Certified ecological prey-response approximation and corresponding testing obstruction. Increasing the finite-state budget sharply reduces the interval-enclosed response error and drives the lower bound on testing error toward chance.

4. Numerical Methods and Validation

The computational layer is designed to support the analytical claims rather than replace them. All scripts and machine-readable outputs are released in the accompanying benchmark and figure directories.

4.1. Solvers

ODE and latent-state systems are integrated with fourth-order Runge–Kutta. Retarded DDEs use an interpolated-history RK4 solver. Caputo systems use the fractional Adams–Bashforth–Moulton PECE scheme [16]; its scalar implementation is validated against the exact Mittag–Leffler solution.

For the combined fractional-delay model (2.4), we use the same PECE history convolution while evaluating the delayed prey term by interpolation of the already computed trajectory. For $\tau > 0$ all reported grids satisfy $\tau \geq h$, so the delayed value required by the corrector lies in known history. The two limiting cases provide strong numerical gates:

- $\tau = 0$ is delegated exactly to the validated Caputo solver;
- at $\alpha = 1$, the fractional-delay solver converges to the independent DDE RK4 solution.

At $A = 0.25$, $\tau = 0.35$, the maximum absolute discrepancy between the $\alpha = 1$ fractional-delay solver and the DDE solver is 6.75×10^{-6} at $N = 200$, 1.68×10^{-6} at $N = 400$, and 4.20×10^{-7} at $N = 800$. For the representative fractional-delay cell $(\alpha, \tau) = (0.85, 0.35)$, the max difference against the $N = 800$ solution drops from 2.18×10^{-3} at $N = 200$ to 8.05×10^{-4} at $N = 400$.

4.2. Inputs, observations, and metrics

Six bounded inputs are compared at the common peak-amplitude budget $\|u\|_\infty \leq 0.10$: pulse, multiscale pulse train, sinusoid, multisine, chirp, and pseudorandom binary sequence (PRBS). Observations use prey, predator, or both channels. Numerical safety is reported separately from solver non-divergence using the minimum Allee margin

$$\min_t (x(t) - A)$$

and the rate of threshold crossing.

For fractional-delay experiments we additionally report (i) trajectory distance to the pure Caputo limit, (ii) trajectory distance to the pure DDE limit, (iii) predator peak time and amplitude, and (iv) a robust bridge-discrimination score defined as the minimum of the distances to the two limiting explanations.

4.3. Validation matrix and numerical resolution

The main numerical claims are associated with explicit validation gates rather than a single solver run. Table 3 collects the convergence checks used in the released package. The scalar Caputo PECE implementation is compared with an exact Mittag–Leffler solution. The combined fractional-delay implementation is checked independently at the integer-order boundary against the DDE RK4 solver, and an interior fractional-delay cell is subjected to self-convergence. The error decreases monotonically in every reported sequence.

Table 3. Numerical validation gates. The Caputo column uses the exact Mittag–Leffler reference; the fractional-delay boundary uses an independent DDE RK4 solution.

Gate	grid size	Reference	max. abs. discrepancy
Caputo PECE	$N = 250$	exact Mittag–Leffler	8.70×10^{-4}
Caputo PECE	$N = 500$	exact Mittag–Leffler	4.64×10^{-4}
Caputo PECE	$N = 1000$	exact Mittag–Leffler	2.52×10^{-4}
Fractional-delay, $\alpha = 1$	$N = 200$	DDE RK4	6.75×10^{-6}
Fractional-delay, $\alpha = 1$	$N = 400$	DDE RK4	1.68×10^{-6}
Fractional-delay, $\alpha = 1$	$N = 800$	DDE RK4	4.20×10^{-7}
Fractional-delay, $(\alpha, \tau) = (0.85, 0.35)$	$N = 200$	$N = 800$	2.18×10^{-3}
Fractional-delay, $(\alpha, \tau) = (0.85, 0.35)$	$N = 400$	$N = 800$	8.05×10^{-4}

The validation strategy is deliberately heterogeneous. An agreement between two implementations that share the same discretization can hide a common coding error; therefore the most important boundaries are checked against a mathematically different reference. At $\tau = 0$, the fractional-delay code delegates to the already validated Caputo implementation rather than duplicating the same calculation. At $\alpha = 1$, by contrast, the memory convolution collapses to an integer-order problem and the output is compared with the separate interpolated-history DDE RK4 path. The two checks test opposite boundaries of the combined model.

4.4. Experimental grid and reproducibility

The benchmark is organized around a small number of interpretable axes rather than unconstrained parameter search. The common peak actuator budget is $\|u\|_\infty \leq 0.10$. The original four-class stable benchmark varies Allee level, fractional order, SNR, observation channel, and waveform family; each cell is replicated 200 times. The dedicated

Table 4. Main numerical-design axes. The fractional-delay campaign is intentionally targeted and is separate from the larger four-class benchmark.

Component	Values / protocol
Peak input budget	$\ u\ _\infty \leq 0.10$ for all six waveform families
Four-class stable benchmark	ODE, Caputo, retarded DDE, latent-3; $A \in \{0.20, 0.25\}$; 200 replicates per cell
Fractional orders in benchmark	$\alpha \in \{0.70, 0.85, 0.90, 0.95\}$ where applicable
Observation channels	prey only, predator only, both species
Noise regimes	high, medium, and low SNR
Fractional-delay design grid	$A = 0.25$, $\alpha \in \{0.75, 0.85, 0.95\}$, $\tau \in \{0.15, 0.30, 0.45\}$
Fractional-delay bridge grid	$\alpha = 0.70, 0.75, \dots, 1.00$ and $\tau = 0, 0.10, \dots, 0.50$
Simulation horizon	$T = 12$ for the principal trajectory/design comparisons

fractional-delay campaign is smaller and targeted: it uses $A = 0.25$, $\alpha \in \{0.75, 0.85, 0.95\}$, $\tau \in \{0.15, 0.30, 0.45\}$ for the design comparison, and a denser (α, τ) grid for the bridge plots. Table 4 summarizes the main numerical axes.

All quantities used in the figures are generated from machine-readable JSON or NPZ outputs. The fractional-delay script writes the solver validation sequence, the design ranking, and the bridge arrays together with the four figures. This separation between simulation generation and paper rendering is useful for audit: plots can be regenerated without rerunning the large four-class benchmark, while the archived benchmark statistics can be checked independently.

5. Frequency-Domain and Finite-Horizon Evidence

The structural results are visible before any large benchmark is run. Figure 6 shows transfer magnitudes on the prey channel at $A = 0.25$. Figure 7 shows the phase separation and the high-frequency convergence to $-\alpha\pi/2$ on collocated channels.

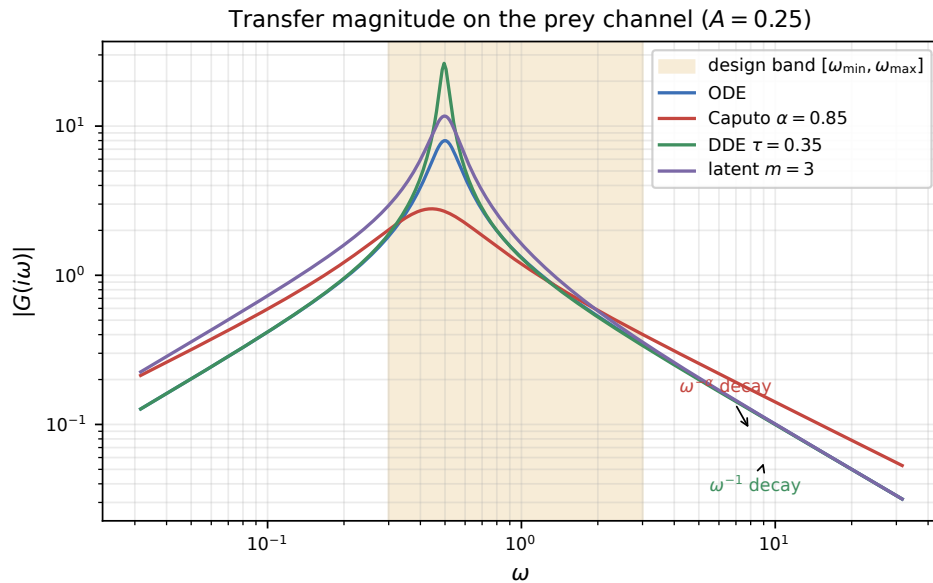


Figure 6. Transfer magnitude on the prey channel for the four memory mechanisms at $A = 0.25$. The fractional response separates across the design band and approaches its noninteger high-frequency decay.

5.1. From exact signatures to finite-horizon ambiguity

The two frequency-domain panels should be read together. The noninteger high-frequency decay and phase give a structural signature of Caputo dynamics, but an experiment does not observe an asymptotic half-line of frequencies. It observes a finite bandwidth over a finite time window and with finite noise. A sufficiently flexible latent model can therefore mimic the relevant part of the response without being analytically identical to the fractional transfer. This is why the structural theorem and the finite-horizon impossibility theorem are complementary rather than competing statements.

The ecological prey-response calculation makes the same point directly in the controlled population channel. For

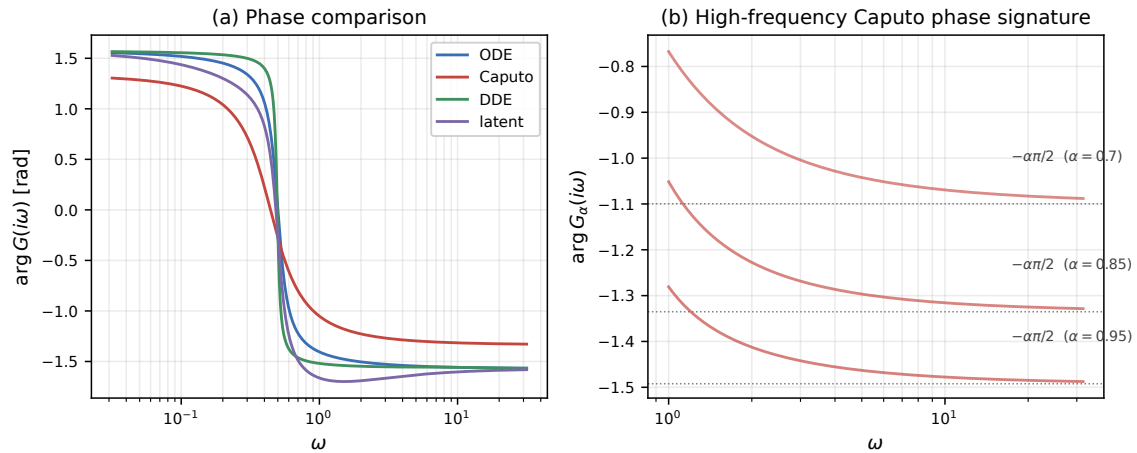


Figure 7. Phase evidence for the fractional signature. Left: Caputo, ODE, DDE, and finite latent responses across frequency. Right: the Caputo phase converges to the scale-free asymptote $-\alpha\pi/2$.

$\alpha = 0.85$ and $T = 12$, Table 5 reports certified L^1 response bounds together with the corresponding Pinsker lower bounds under the declared pulse and $\sigma = 0.10$. The approximation improves by almost two orders of magnitude between $m = 4$ and $m = 32$, and the lower bound correspondingly approaches the equal-prior chance level $1/2$.

Table 5. Certified finite-state approximation of the fractional prey response and induced testing lower bound at $\alpha = 0.85$, $T = 12$, pulse $\|u\|_2 \leq 0.120$, and $\sigma = 0.10$.

m	$\hat{E}_m^{\text{state}}$	$A = 0.25$		$A = 0.30$	
		P_e^*	lower bound	$\hat{E}_m^{\text{state}}$	P_e^*
4	0.5334	0.340		0.4706	0.359
8	0.1254	0.462		0.0898	0.473
16	0.0273	0.492		0.0204	0.494
32	0.0069	0.498		0.0066	0.498

The table is deliberately a one-sided statement. A large lower bound on P_e^* proves that the specified experiment cannot reliably distinguish the focal fractional response from at least one rival in the admitted finite-state class; the converse is not claimed. Cells in which the bound is small are therefore *inconclusive*, not automatically discriminable. This interpretation is preserved in the atlas of Figure 8.

The finite-horizon approximation results then explain why frequency-domain separation does not automatically translate into easy model selection. The safe memory-discrimination atlas in Figure 8 displays the protocol-restricted lower bound on testing error across fractional order, Allee parameter, and latent complexity. Cells outside the focal Matignon-stable regime are explicitly excluded from the headline interpretation.

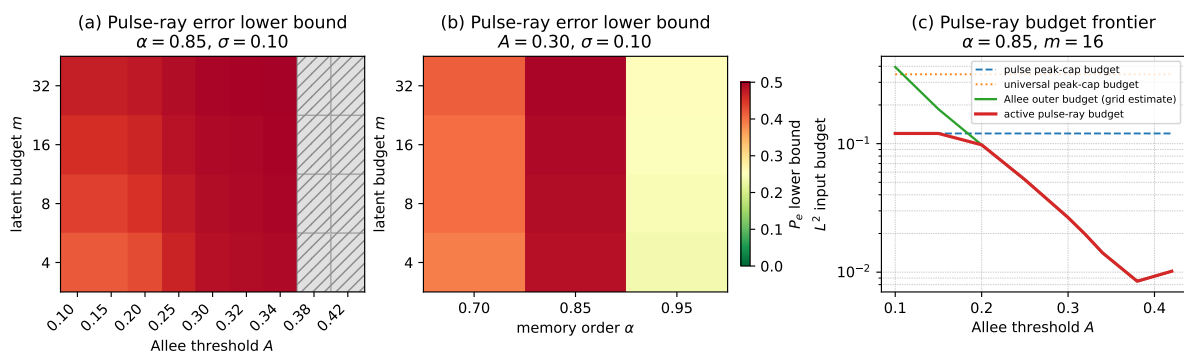


Figure 8. Protocol-restricted memory-discrimination atlas. Increasing latent complexity and moving toward weakly separated fractional regimes raise the lower bound on model-selection error. Hatched cells fall outside the focal Caputo stable regime.

6. Fractional-Delay Model: Dedicated Simulation Study

A natural robustness question motivates a dedicated numerical study of a mechanism carrying *both* fractional memory and explicit delay. Equation (2.4) is useful because its boundaries are exactly the two previously studied mechanisms: $\tau = 0$ gives the pure Caputo model, while $\alpha = 1$ gives the DDE benchmark. The interior of the (α, τ) plane therefore tests whether the two effects reinforce, compensate, or create ambiguous response signatures.

The purpose of this section is not to claim a new general theory of fractional delay equations. It is an application-oriented stress test of the discrimination framework. The same strong-Allee ecology, prey perturbation channel, amplitude budget, and observation variables are kept fixed while two memory coordinates are varied. This isolates a practical concern that is difficult to see in pairwise model comparisons: a real system may contain both a broad hereditary response and a physiologically meaningful delay, so a binary “fractional versus delay” interpretation can be misspecified even when each limiting model fits part of the trajectory.

6.1. Transient trajectories

Figure 9 fixes $A = 0.25$, $\alpha = 0.85$, and applies the same prey pulse while increasing τ . The prey response changes moderately, whereas the predator response is visibly shifted and its late transient becomes increasingly delay-sensitive. Importantly, the trajectories remain above the Allee threshold for this pulse experiment.

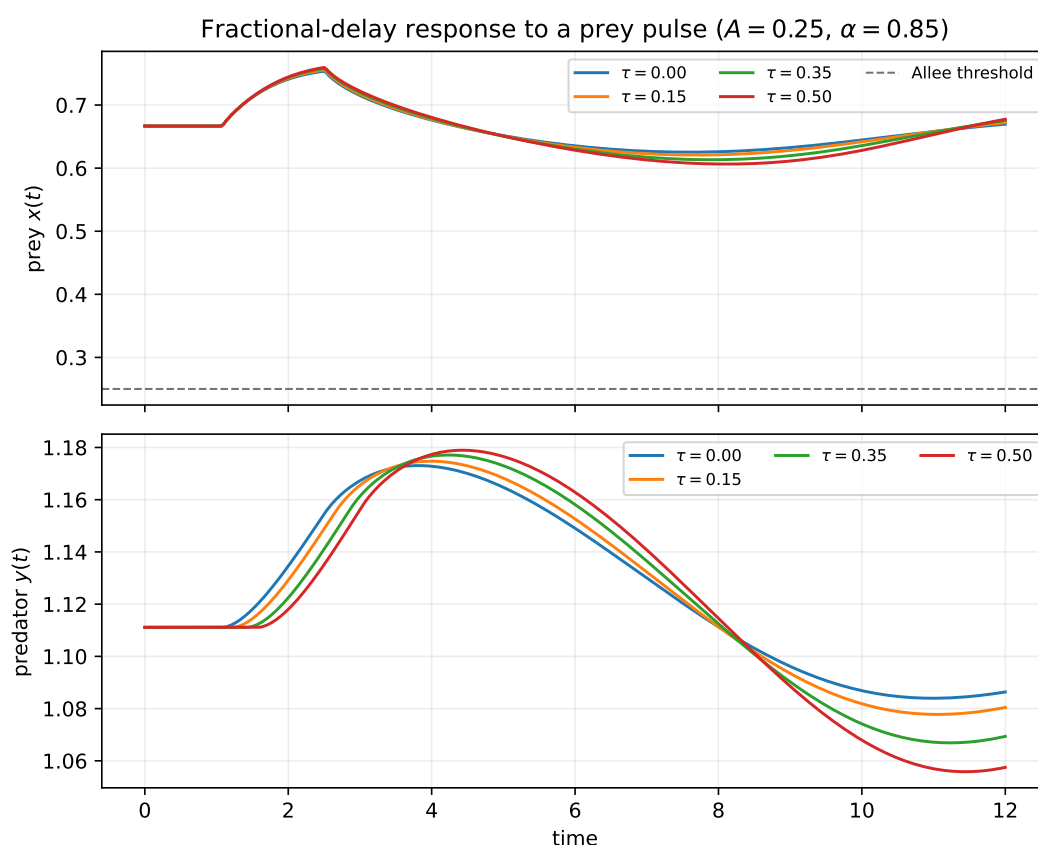


Figure 9. Combined fractional-delay response to the same prey pulse at $A = 0.25$ and $\alpha = 0.85$. Increasing delay shifts the predator response and changes the late transient while preserving the fractional-memory relaxation pattern.

6.2. Fractional-order/delay parameter plane

Figure 10 sweeps $\alpha \in [0.70, 1]$ and $\tau \in [0, 0.5]$. The left boundary has zero distance to the pure Caputo model by construction; the top boundary has zero distance to the pure DDE model. The interior quantifies how rapidly the combined mechanism departs from either limiting explanation. The safety panel shows that the same delay can be benign at lower α but erode the Allee margin as the system approaches the integer-order limit.

Table 6 gives representative numerical values from the bridge grid. For fixed α , increasing τ moves the trajectory monotonically farther from the pure Caputo boundary. At the same time, the trajectory remains substantially separated

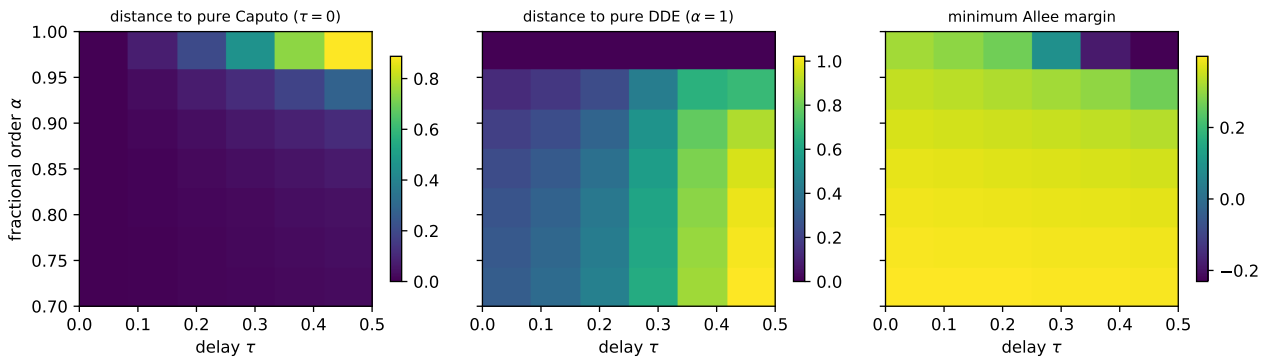


Figure 10. Fractional-delay bridge over (α, τ) . Left: trajectory distance to the pure Caputo limit. Center: distance to the pure DDE limit. Right: minimum Allee margin under the same pulse. The interior of the parameter plane contains responses that are not well represented by either limit alone.

from the pure DDE boundary for the displayed cells. The effect is especially strong near $\alpha = 0.95$: the distance to the Caputo limit grows from 0.0299 at $\tau = 0.10$ to 0.2805 at $\tau = 0.50$, while the minimum Allee margin falls from 0.3345 to 0.2629. This is a direct numerical example of a region in which explicit delay increases both identifiability and ecological risk.

Table 6. Representative cells from the fractional-delay bridge at $A = 0.25$ under the pulse experiment. Distances are trajectory L^2 distances to the corresponding limiting models.

α	τ	distance to Caputo	distance to DDE	min. $[x(t) - A]$
0.75	0.10	0.0063	0.3426	0.3917
0.75	0.30	0.0182	0.6145	0.3887
0.75	0.50	0.0315	1.0086	0.3850
0.85	0.10	0.0103	0.2900	0.3724
0.85	0.30	0.0334	0.5683	0.3655
0.85	0.50	0.0625	0.9606	0.3566
0.95	0.10	0.0299	0.1645	0.3345
0.95	0.30	0.1142	0.4330	0.3123
0.95	0.50	0.2805	0.6875	0.2629

6.3. Which input exposes the combined mechanism?

For $\alpha \in \{0.75, 0.85, 0.95\}$ and $\tau \in \{0.15, 0.30, 0.45\}$, we score each input by the mean of the minimum trajectory distance to the pure Caputo and pure DDE explanations. Figure 11 shows that multisine excitation gives the largest bridge-discrimination score, but it crosses the Allee threshold in one third of the tested cells. PRBS, sinusoid, and chirp cross in every tested cell; pulse and multiscale excitation remain above threshold but are less discriminating. This reproduces the global safety–informativeness tension on a model that contains both kinds of memory simultaneously.

The complete ranking is reported in Table 7. The score is the mean, over the nine (α, τ) cells, of the minimum trajectory distance to the pure Caputo and pure DDE limits. This deliberately penalizes inputs that separate the combined response from only one of the two explanations. The second and third columns show that the highest scores are achieved by sustained/broadband perturbations, whereas the two transient families maintain positive Allee margins throughout the tested grid.

6.4. Delay sweep

The delay sweep in Figure 12 makes the mechanism interpretable. Increasing τ shifts the predator peak later for every tested fractional order and increases its peak amplitude modestly. At the same time the minimum Allee margin decreases, especially near $\alpha = 1$. Thus the delay is visible not merely as an additional fitting parameter but as a reproducible shift in response timing and safety margin.

For example, at $\alpha = 0.85$ the predator peak moves from $t \approx 3.81$ at $\tau = 0$ to $t \approx 4.50$ at $\tau = 0.55$, while the minimum Allee margin decreases from 0.3759 to 0.3548. At $\alpha = 0.95$ the same delay change moves the peak from

Table 7. Fractional-delay design stress test at $A = 0.25$. The score averages the minimum distance to the Caputo and DDE limits over $\alpha \in \{0.75, 0.85, 0.95\}$ and $\tau \in \{0.15, 0.30, 0.45\}$.

Input	mean bridge score	worst Allee margin	crossing fraction
multisine	0.1332	-0.2103	0.333
PRBS	0.0909	-0.2814	1.000
sinusoid	0.0894	-0.2777	1.000
chirp	0.0861	-0.2763	1.000
pulse	0.0627	0.2770	0.000
multiscale	0.0338	0.3171	0.000

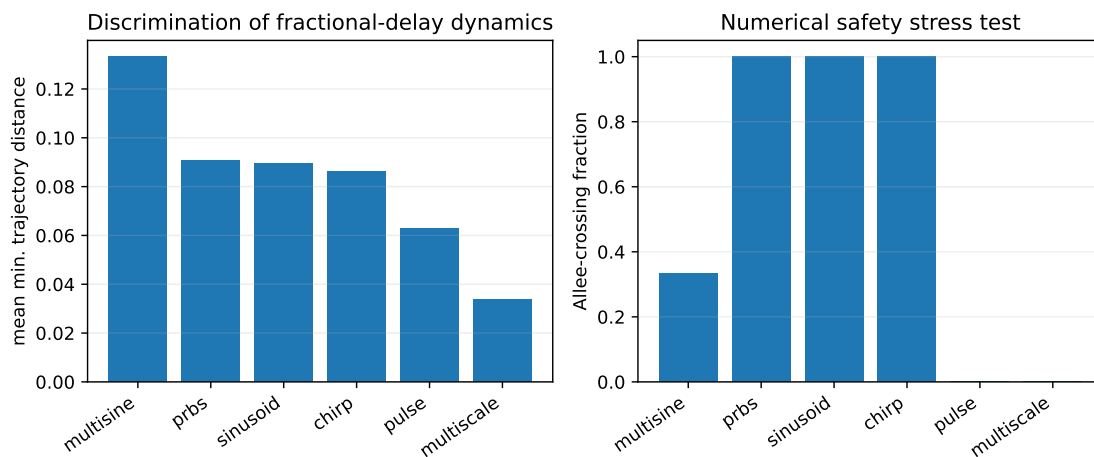


Figure 11. Input-design stress test for fractional-delay dynamics. Left: mean minimum trajectory distance from the two limiting explanations (pure Caputo and pure DDE). Right: numerical Allee-crossing fraction on the same (α, τ) grid. Broadband inputs reveal the combined mechanism more strongly but are substantially less safe.

$t \approx 4.16$ to 4.89 and reduces the Allee margin much more strongly, from 0.3434 to 0.2384 . Table 8 provides the endpoint comparison for all three fractional orders.

Table 8. Endpoint summary of the delay sweep under the pulse experiment. Values compare $\tau = 0$ with $\tau = 0.55$.

α	τ	predator peak time	predator peak	min. Allee margin
0.75	0	3.343	1.1634	0.3934
0.75	0.55	4.029	1.1681	0.3843
0.85	0	3.814	1.1724	0.3759
0.85	0.55	4.500	1.1788	0.3548
0.95	0	4.157	1.1838	0.3434
0.95	0.55	4.886	1.1925	0.2384

The trend also clarifies why dense observation near the predator peak is useful. Delay is expressed strongly through timing, while fractional memory is expressed over a broader relaxation tail. A sampling scheme that observes only a short window around the initial prey perturbation can therefore miss both signatures: it may not wait long enough for the delayed predator response, and it may not observe long enough to resolve the fractional relaxation.

7. Experiment Design and Safety

The analytical design problem is summarized in the main paper and proved in full in the Supplementary Material. For two Gaussian linearized models with common covariance, the pairwise information criterion can be written

$$D_{ij}(u) = \frac{1}{2} \langle u, Q_{ij} u \rangle.$$

Under an energy constraint, the unconstrained optimum is the leading eigenfunction of Q_{ij} ; for a finite robust frequency design, an optimum can be represented with finite frequency support. These standard design consequences are used here only to motivate the six implementable waveform families in Figure 13.

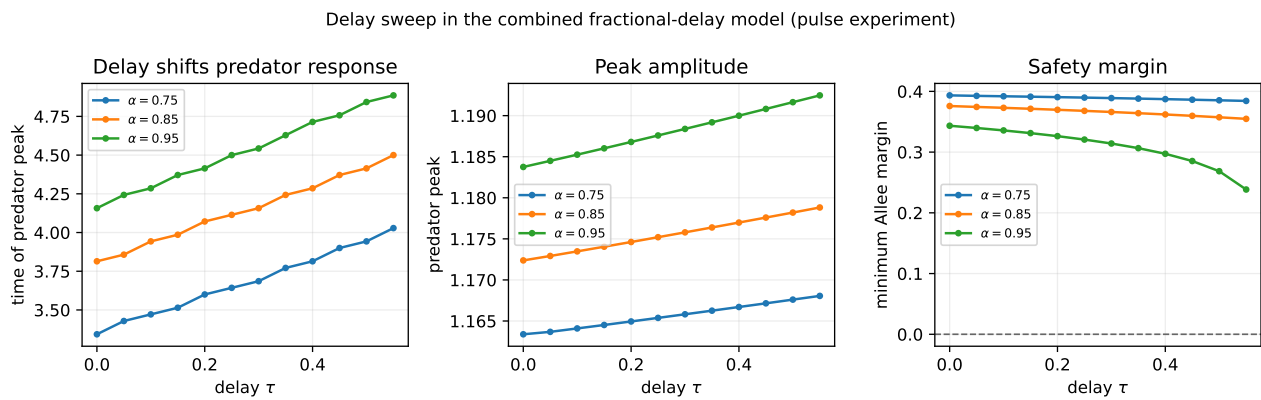


Figure 12. Delay sweep under the pulse experiment. The predator peak occurs later as τ grows, peak amplitude increases moderately, and the numerical Allee margin decreases. These trends are consistent across the three displayed fractional orders.

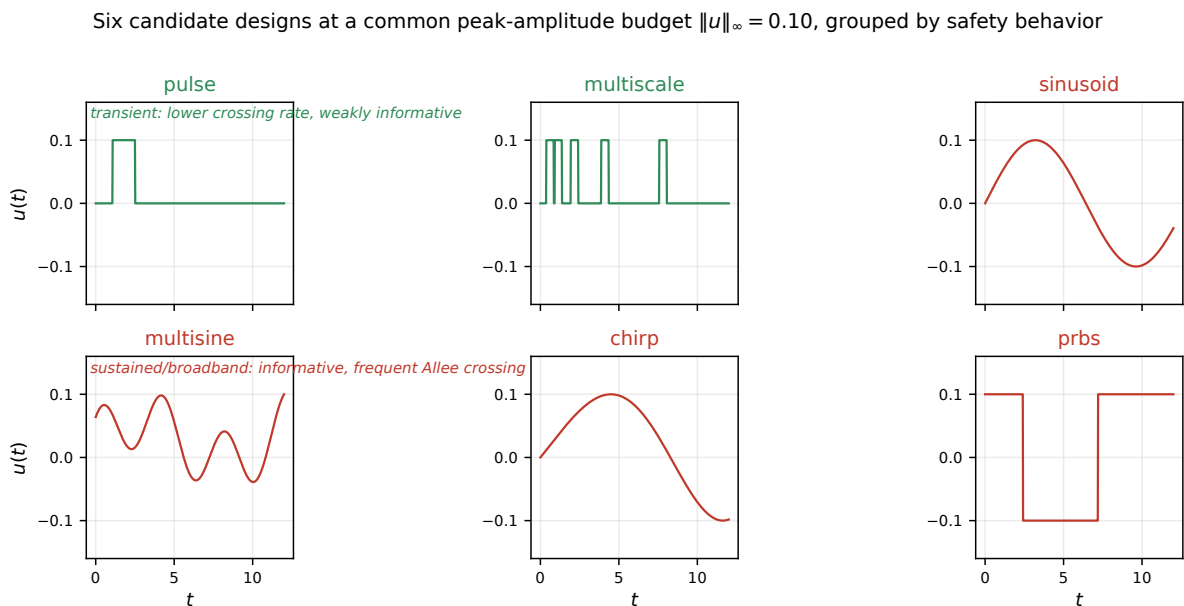


Figure 13. Six candidate perturbations at the common peak-amplitude budget. Transient and broadband designs probe the memory law differently and exhibit markedly different safety behavior in nonlinear simulations.

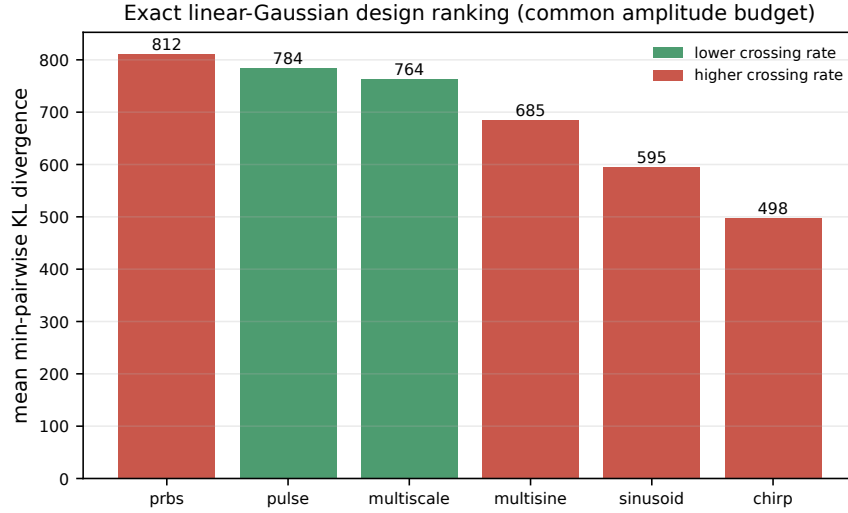


Figure 14. Linear-Gaussian design ranking by mean minimum pairwise KL divergence. This ranking ignores nonlinear state safety and therefore acts as an upper-level information benchmark rather than a final experimental recommendation.

7.1. Strong-Allee safety

A one-sided Allee constraint requires

$$x(t) > A + \eta.$$

For the linearized system a sufficient pulse-amplitude envelope follows from the prey impulse gain; for the nonlinear system the strict inward-pointing rectangle theorem and its face inequalities are given in the Supplementary Material. Crucially, state safety alone does not impose a universal bound on input energy without a physical waveform/bandwidth class, because high-frequency inputs may be strongly attenuated by a stable strictly proper channel. This is why all quantitative testing bounds in the main paper are tied to an explicit actuator or protocol class.

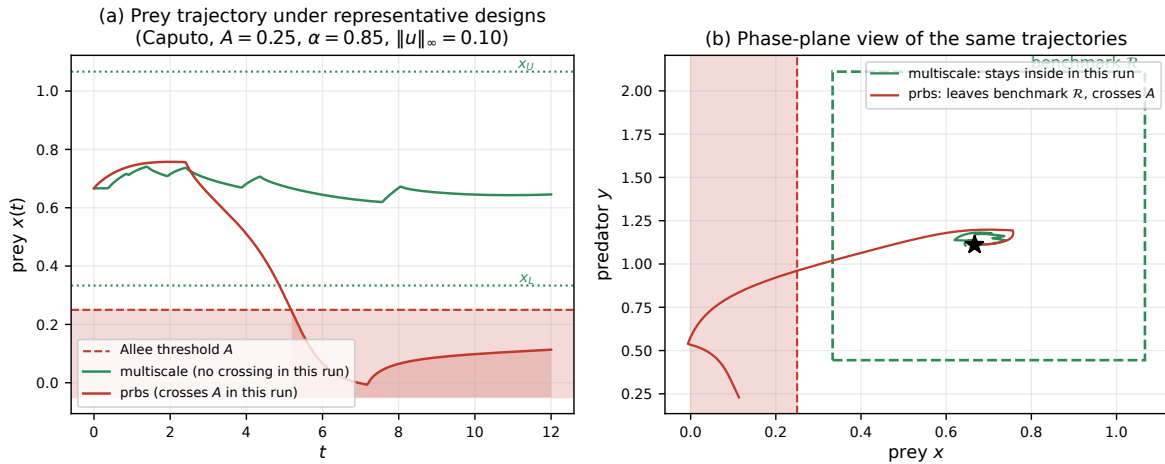


Figure 15. Representative nonlinear safety geometry. A multiscale transient remains above the Allee threshold in the shown run, while PRBS crosses it. The benchmark rectangle is a diagnostic region; rigorous positive invariance requires the strict face conditions stated in the Supplementary Material.

8. Large-Scale Model-Discrimination Benchmark

The calibrated benchmark compares ODE, Caputo, retarded DDE, and finite latent mechanisms at matched operating conditions. The primary stable-regime task uses $A \in \{0.20, 0.25\}$; a separate $A > 2/7$ task probes the easier stability-verdict problem. The decision rule is BIC, not Bayesian model evidence.

8.1. Benchmark design and decision rule

The purpose of the benchmark is not to find a single best-fitting model on one synthetic trajectory. Instead, it asks whether the same qualitative design conclusions survive across changes in memory strength, observation noise, observation channel, and input waveform. For each generated trajectory, each candidate model is fitted under the same observation protocol and compared using the Bayesian information criterion (BIC). The benchmark therefore evaluates a reproducible information-criterion decision rule; it does not report posterior model probabilities or Bayesian evidence.

The stable-regime benchmark contains 1512 factorial cells, with 200 stochastic replicates per cell and no solver divergences. A separate linear-Gaussian factorial contains 1458 cells and is used only as an information-ranking reference. Keeping these two layers separate is important: the linear ranking indicates where the models differ most under local approximations, whereas the nonlinear benchmark includes the strong-Allee geometry and can therefore reverse the practical recommendation when an informative input becomes unsafe.

The full-factorial benchmark contains 200 replicates per cell and no solver divergences. In the primary four-class stable task, macro-averaged accuracy is 0.537 (chance 0.25). The design ranking is

$$\begin{aligned} \text{PRBS (0.584)} &\approx \text{multisine (0.583)} > \text{sinusoid (0.534)}, \\ &> \text{chirp (0.507)} > \text{pulse (0.414)} > \text{multiscale (0.324)}. \end{aligned}$$

Discrimination degrades sharply as $\alpha \rightarrow 1$: accuracy falls from approximately 0.70 at $\alpha = 0.70$ to 0.31 at $\alpha = 0.95$. It is also noise-limited: 0.76 at high SNR, 0.44 at medium SNR, and 0.27 at low SNR. Observing both species performs best, although prey-only observation remains structurally sufficient.

Table 9 shows the main stratified accuracies. The largest degradation is associated with the two mechanisms becoming intrinsically similar ($\alpha \rightarrow 1$) and with decreasing SNR. Observation design has a smaller but still meaningful effect: both-species observation gives the highest mean accuracy, prey-only is close behind, and predator-only loses information about the directly actuated state.

Table 9. Stable-regime four-class BIC accuracy stratified by fractional order, SNR, and observation channel.

Axis	level	accuracy	Axis	level	accuracy
α	0.70	0.702	SNR	high	0.756
	0.85	0.557		medium	0.444
	0.90	0.467		low	0.273
	0.95	0.312	channel	prey	0.516
				predator	0.427
				both	0.530

The class-wise statistics provide additional interpretation. ODE recall is very high (0.972), but ODE precision is only 0.227: when the excitation is weak, several subtle memory mechanisms are absorbed into the simpler ODE class. Caputo selection shows the opposite pattern, with recall 0.524 but precision 0.984; when the benchmark has enough evidence to select Caputo, that decision is rarely a false positive. DDE recall/precision are 0.410/0.731, and latent-3 recall/precision are 0.242/0.848. Thus the main failure mode is not indiscriminate confusion; it is conservative collapse toward the simpler model when the available excitation does not reveal enough memory structure.

8.2. Safety-informativeness trade-off

Numerical state safety is reported separately from information. At the common peak-amplitude budget, multiscale excitation has zero Allee crossings in the stable benchmark but the weakest discrimination. PRBS and sinusoidal excitation have the strongest discrimination among the tested families but cross the Allee threshold in essentially all stable-regime cells. Multisine occupies an intermediate position: high discrimination with a lower, but still non-negligible, crossing rate.

Table 10 reports the archived nonlinear safety statistics. The ranking makes the trade-off concrete. Multiscale excitation is the safest tested family in the stable benchmark and has a positive minimum Allee margin of 0.292, but it also has the lowest classification accuracy. PRBS reaches the highest accuracy and crosses the threshold in every stable-regime cell. The pulse is a useful compromise: its crossing rate is only 0.071, but its accuracy remains substantially below the broadband designs. Multisine is the most interesting intermediate design, with nearly the same discrimination as PRBS but a crossing rate of 0.357 rather than 1.0.

This table also explains why the linear-Gaussian information ranking cannot by itself select a final experiment. In the linear calculation the input is judged by separation of response operators. In the nonlinear ecology the same input

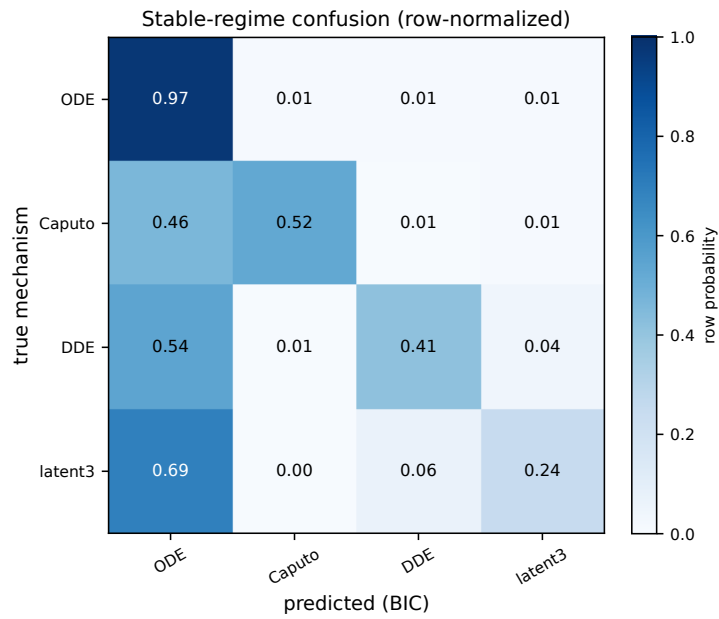


Figure 16. Row-normalized confusion matrix for the stable four-class task. The simplest ODE mechanism is frequently selected under weak excitation, whereas Caputo predictions are highly precise when selected.

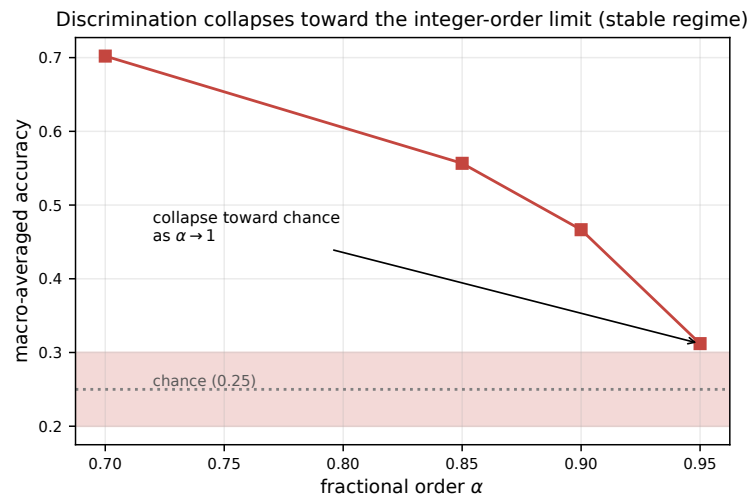


Figure 17. Model-selection accuracy versus fractional order. As α approaches the integer-order limit, the memory mechanisms become increasingly difficult to distinguish.

Table 10. Nonlinear stable-benchmark trade-off at the common peak-amplitude budget. Safety values are numerical benchmark diagnostics, not invariant-set certificates.

Input	macro accuracy	Allee crossing rate	minimum Allee margin
PRBS	0.584	1.000	-0.3139
multisine	0.583	0.357	-0.2405
sinusoid	0.534	1.000	-0.3058
chirp	0.507	0.929	-0.3012
pulse	0.414	0.071	-0.1207
multiscale	0.324	0.000	0.2921

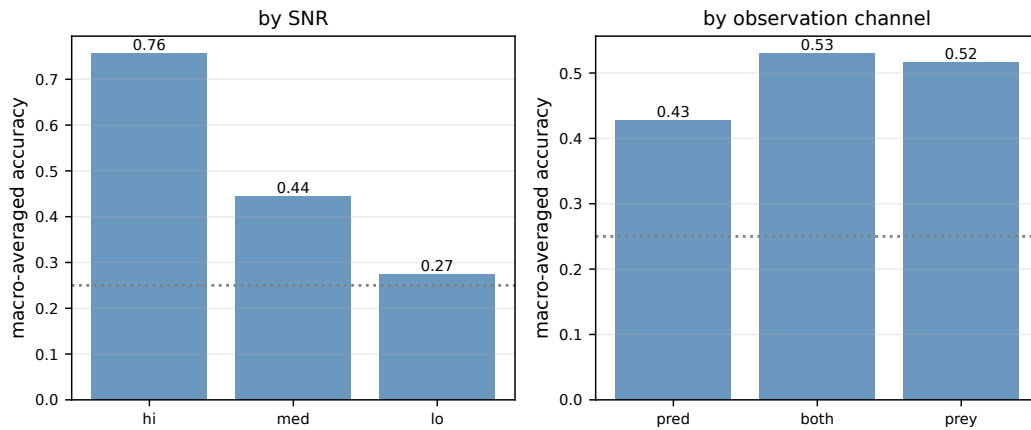


Figure 18. Accuracy stratified by SNR and observation channel. The benchmark is strongly noise-limited; both-species observation provides the highest average accuracy.

can move the system into a different basin, at which point the perturbation is no longer a local identification experiment. Safety must therefore be evaluated on the nonlinear state trajectory, not inferred from the linear information score.



Figure 19. Headline benchmark trade-off. Broadband/sustained designs move upward in discrimination but also toward larger Allee-crossing rates; transient designs occupy the lower-crossing but less-informative region.

9. Prospective Experimental Interpretation

The numerical results translate into a practical microcosm protocol. A useful first experiment should (i) operate in a stable strong-Allee regime, (ii) apply a bounded prey perturbation, (iii) observe both populations when feasible, and (iv) combine a safe transient excitation with a second, more broadband perturbation only if the first experiment leaves sufficient safety margin. Laboratory predator–prey microcosms provide a natural setting for such controlled perturbations [2, 7, 22].

The fractional-delay study adds a concrete diagnostic: delayed feedback should shift the timing of the predator peak, whereas decreasing fractional order primarily modifies the long transient and phase response. A design that records dense observations around the predator peak and continues sufficiently long to resolve the fractional tail is therefore more informative than uniform sparse sampling alone.

A Bayesian sequential layer is specified in the Supplementary Material but is not executed here. The present paper

deliberately keeps the main evidence reproducible with deterministic solvers, interval enclosures, and BIC-based simulation benchmarks.

A two-stage prospective protocol

The numerical results suggest a staged experiment rather than a single maximally informative perturbation. Stage 1 uses a transient input with a low observed crossing rate (pulse or multiscale) to estimate the local response scale and verify that the realized trajectory remains comfortably above the Allee threshold. Stage 2 is conditional: only if the first experiment leaves adequate margin should a more informative multisine-type excitation be applied. This sequencing is not claimed to be Bayes-optimal; it is a practical interpretation of the observed safety–information frontier.

Table 11. Prospective microcosm workflow suggested by the simulation results.

Stage	Perturbation	Primary measurements	Decision / purpose
0	no perturbation	baseline prey/predator variability	estimate noise and confirm operating point
1	pulse or multiscale	dense early prey observations; predator peak timing; minimum prey level	verify local response and available safety margin
2	multisine if margin permits	both species over the full transient and relaxation tail	maximize separation of memory mechanisms
3	optional repeat / altered horizon	same channels, replicated under matched conditions	test robustness of the selected mechanism and estimate variability

Three measurement features are especially informative. First, the directly actuated prey channel carries the strongest early signal and should be sampled densely immediately after the perturbation. Second, the delayed trophic pathway is expressed through the timing of the predator response, so observations should be concentrated around the anticipated predator peak rather than uniformly diluted over the entire window. Third, fractional memory is most visible in the slower relaxation tail; the observation horizon should therefore extend well beyond the peak. The combined design is naturally nonuniform: dense around the forcing and predator peak, then sparser but longer during relaxation.

The present simulations use dimensionless state variables and a normalized perturbation budget. A physical implementation would first map these quantities to organism counts, biomass, or resource manipulation units and would have to define an operational safety margin above the empirical Allee threshold. That calibration step is application-specific and is intentionally not inferred from simulation alone.

10. Discussion

The restructured results support three conclusions.

First, *structural identifiability and practical discrimination are different questions*. The Caputo transfer has an exact noninteger signature, yet finite latent models approximate the same response increasingly well on finite horizons. The testing bounds quantify this gap and explain why a flexible latent alternative can remain difficult even when the underlying model classes are analytically distinct.

Second, *delay and fractional memory should not be treated as mutually exclusive numerical explanations*. The combined fractional-delay simulations show a continuous bridge between the pure Caputo and pure DDE limits. Delay systematically shifts the predator response, while the fractional order controls the broader relaxation structure. The interior of the (α, τ) plane contains dynamics that can be poorly represented by either pure limit, which is precisely why direct numerical stress testing is valuable.

Third, *the most informative perturbation is not automatically the safest*. The same conclusion appears in the original four-class benchmark and in the new fractional-delay experiments. Broadband inputs enlarge response differences but more often drive prey below the strong-Allee threshold. This tension is ecological, not merely numerical: crossing A changes the long-term basin structure and invalidates an experiment designed only around local identifiability.

Limitations. The certified ecological response theorem is linearized and prey-channel specific; a full-state hierarchy-wide safety closure remains open. The fractional-delay study is numerical and uses a constant equilibrium history, one delayed trophic pathway, and a targeted parameter grid. The benchmark uses BIC rather than fully Bayesian evidence and does not execute adaptive design. These restrictions are stated explicitly to separate proved results from computational evidence.

Relation to the previously submitted companion paper. The present manuscript shares the strong-Allee ecological backbone and locked baseline parameterization with Ref. [12], but it does not repeat that paper’s research question or headline results. Ref. [12] concerns computer-assisted certification of Caputo equilibria, the Matignon stability boundary, extinction/boundedness, and the diagnostic failure of a forward-Euler surrogate. By contrast, the present work studies active discrimination among competing memory mechanisms. The structural transfer separation, finite-horizon latent approximation hierarchy, testing obstruction, certified prey-response approximation, six-family input-design study, four-class discrimination benchmark, and combined fractional-delay simulations are not results of Ref. [12]. The inherited stability facts are used only to select admissible operating regimes and are explicitly attributed where they enter.

What is genuinely added by the present paper. The reuse of the same ecological backbone is a design choice, not a duplication of the previous study. Holding the ecology fixed makes it possible to attribute differences in the simulated observations to the memory law and to the experiment rather than to a changed population model. The new paper therefore starts where the companion paper stops. Once an operating point and stability region have been certified, one must still decide whether observed transients reveal a fractional mechanism uniquely, whether a delayed or finite latent model can mimic them, and which perturbations can separate those alternatives without leaving the safe ecological regime. These are distinct inferential questions, and they require different mathematics and different computations.

The contrast also explains why the figures are different. The companion paper’s central visual object is a certified (A, α) stability atlas. In the present manuscript, the central visuals concern response separation, approximation error, waveform design, BIC confusion, fractional-delay trajectories, and the safety–informativeness frontier. The common model schematic is used only to anchor the reader in the shared ecology; none of the companion paper’s certificate atlas or certificate-count results is reused as evidence for the new discrimination claims.

Relation to broader prior work. General safe information limits and safe active model discrimination are already established [15, 19]; the contribution here is the interaction of finite-horizon fractional approximation, bounded latent complexity, strong-Allee ecological geometry, and direct numerical stress tests of the combined fractional-delay mechanism. The paper therefore does not claim novelty for safe experimental design in the abstract.

Implications for experimental identification. The experiments suggest that model selection should be treated as a controlled intervention problem rather than a passive curve-fitting exercise. With weak excitation, the four candidate mechanisms can produce trajectories that are close enough that BIC often defaults to the simpler ODE explanation. Increasing excitation improves separation but can move the system toward or across the Allee threshold. The combined fractional-delay study adds a second warning: even if a pure Caputo model and a pure DDE are individually distinguishable, an interior model containing both effects may occupy a response region that neither pure limit represents well. Consequently, experimental design should test composite alternatives rather than only the two endpoints one at a time.

A useful practical objective is therefore not “maximize information” in isolation, but maximize information subject to a trajectory-level safety margin and a declared rival-model complexity budget. The latter is essential because an unrestricted finite latent hierarchy can approximate fractional memory arbitrarily well on a fixed finite horizon. Reporting the allowed latent dimension, delay range, forcing bandwidth, and observation horizon is thus part of the scientific specification of the discrimination experiment, not a secondary implementation detail.

11. Conclusion

We developed a compact theory-and-simulation framework for discriminating ecological memory mechanisms under strong-Allee safety constraints. Four analytical results are retained in the main text: structural separation, finite-horizon latent approximation, a complexity-dependent testing obstruction, and an interval-certified finite-state approximation of the ecological prey response. The numerical evidence is substantially broader: frequency-response diagnostics, a dedicated combined fractional-delay simulation study, six perturbation families, safety metrics, and a large four-class discrimination benchmark.

The main empirical message is consistent across these layers. Fractional memory is structurally distinct, but finite experiments may not separate it from delayed or sufficiently rich latent alternatives. Increasing excitation improves discrimination, yet the strongest inputs are also the most likely to cross the Allee threshold. The appropriate next

experimental question is therefore not simply *which model fits best*, but *which safe perturbation makes the competing memory mechanisms observably different*.

Author contributions

The author contributed to the conceptualization, methodology, formal analysis, numerical investigation, visualization, and preparation of the manuscript.

Use of AI tools declaration

AI-assisted tools were used during manuscript preparation for language editing, LaTeX formatting, code and documentation assistance, and structural organization. Responsibility for the scientific content and the final submitted version remains with the author.

Conflict of interest

The author declares no conflict of interest.

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